



Editorial

Neuropsychologia special issue editorial: The neural basis of emotion

What is the nature of emotion, and how it is represented in the brain? These questions have long been the subject of a wide variety of philosophical and psychological investigations. A relative newcomer on the scene has been the modern field of affective neuroscience. A rapidly growing body of knowledge about emotion and the brain has emerged in recent years, drawing on an array of modern affective neuroscience methods such as fMRI, ERP, and lesion studies, spanning all levels of brain organization from the single neuron to the entire human brain (Fox et al., 2018). The goal of this special issue is to highlight some of these recent advances in understanding the neural basis of emotion, ranging from studies of emotion representation to studies of emotion-cognition interactions and their implications for clinical disorders. We hope to advance current knowledge in the field by identifying key outstanding questions, common themes, areas of overlap with clinical research, and promising directions for future research. We expect that the papers in this issue will be of interest to a wide basic and clinical neuroscience audience interested in the neural basis of emotion and its translational implications.

The papers in this special issue span a wide range of topics, approaches, and levels of analysis. Three broad, related topics emerge from these papers. The first topic is the brain representation of emotion and brain-behavior relationships. The second topic is interactions between emotion and cognition, including studies of emotional memory, emotion regulation, and the impairing and enhancing effects of emotion on cognitive task performance. The final topic is individual differences in emotion and emotion-related constructs such as personality. The collection includes papers that take a “deep dive” into each these topics, and others that span multiple topics. In the following section we briefly survey the papers in this special issue, highlighting some common themes and outstanding questions.

1. Neural representation of emotion

How are emotions generated in the brain? In an effort towards answering this question, Dubois et al. (2017) present a novel framework for “causal mapping” of emotion networks in the human brain, combining sophisticated fMRI analyses with a procedure that combines intracranial electrical stimulation with concurrent fMRI (es-fMRI). The authors point out that although there is a wealth of correlational data, relatively little is known about how brain structures interact to cause emotions in humans. The novel approach uses two steps: first, a causal discovery algorithm is used to analyze large existing resting-state fMRI datasets and create a network graph that reflects the causal relationships between brain areas. Next, electrical stimulation of one area is used to map networks of evoked brain activity, and these maps can be combined

with the network graphs from the first step to refine, constrain, and test the predictions of the network model. The ultimate goal of this endeavor is to understand the causal role of individual brain regions and networks of regions in generating subjective, behavioral, and physiological aspects of emotion. The use of electrical stimulation of the amygdala closely interleaved with fMRI scanning is a novel approach to map the connectivity of this key affective brain structure. The connectivity findings with es-fMRI are provocative and suggest that the interpretation of some previous functional connectivity studies may need to be modified. In general, this paper outlines a framework for an ambitious research plan that can advance understanding of how emotions are generated from brain networks and eventually guide efforts to modify these networks to treat affective disorders.

Inman et al. (2018) also investigated the effects of amygdala stimulation in humans. However, whereas Dubois et al. intentionally stimulated the amygdala at a level too weak to elicit physiological or subjective emotional effects (focusing instead on the fMRI correlates of stimulation), the main goal of Inman et al.’s study was to elucidate the behavioral and psychophysiological effects of higher levels of amygdala stimulation. A key debate regarding the neural representation of emotion in the human brain is the extent to which individual regions vs. distributed networks play a causal role in generating specific emotions. The amygdala has been implicated in having an important role in both discrete, basic emotions such as fear and in the affective dimension of arousal (both emotion intensity and sympathetic arousal), but causal evidence in humans has been relatively limited and inconsistent. Temporal lobe epilepsy patients undergoing intracranial electrode monitoring were stimulated unilaterally in the amygdala (vs. sham stimulation and control stimulation in the lateral temporal lobe) at a range of amplitude levels. Skin conductance response (SCR), heart rate (HR), and respiratory rate (RR) were simultaneously recorded, and subjective emotional responses were probed after each stimulation period.

Consistent with the amygdala’s role in emotional arousal, amygdala stimulation elicited a rapid and robust dose-dependent increase in skin conductance (a key measure of sympathetic ANS activation). Heart rate also decreased, consistent with a defensive cascade response characteristic of responses seen with aversive stimuli (Bradley et al., 2001), and consistent with a role for the amygdala in both sympathetic and parasympathetic autonomic responses (Beissner et al., 2013; Gianaros and Wager, 2015). Unexpectedly, most of the patients reported feeling no subjective emotional responses despite exhibiting relatively large physiological responses during amygdala stimulation. This dissociation between subjective and physiological responses is worthy of further investigation, and this finding highlights the importance of concurrently

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assessing multiple domains of emotional response. In the two patients who had subjective emotional responses to amygdala stimulation, these responses were sometimes highly intense, most typically manifested as feelings of fear and anxiety. In general, the results of this study suggest that amygdala stimulation elicits responses that have both dimensional and discrete aspects. Applying the causal discovery methods outlined by Dubois et al. would be an excellent next step in this line of research and would help to determine what networks are associated with the physiological and subjective emotional responses to stimulation of the amygdala and other brain regions implicated in affective processing, such as the insula.

As noted previously, a key ongoing debate is the extent to which categorical vs. dimensional emotion constructs are represented in the brain. A related question is how emotions dynamically unfold in time both psychologically and neurophysiologically. The paper by Grootswagers et al. (2017) addresses both of these issues using magnetoencephalography (MEG) and multivariate pattern analysis (MVPA) to examine the relationship between subject ratings of emotional valence and discrete, categorical emotion (i.e., fear, sadness) and whole-brain MEG responses to emotional pictures. Both dimensional and categorical emotion could be discriminated relatively quickly, generally within approximately the first 200–300 milliseconds after picture onset. Different discrete emotions had different temporal dynamics, with the fear-associated pattern detected first and anger appearing last, and the patterns associated with some emotions such as disgust lingered longer than other emotions such as fear. The finding that both dimensional and categorical emotion constructs are mirrored in patterns of distributed brain activity echoes similar findings with fMRI (Chikazoe et al., 2014; Kragel and LaBar, 2016; Saarimäki et al., 2018), and suggests that these representations capture different aspects of the neural representation of emotion. A key direction for further study would be to connect the whole-brain high temporal resolution findings from this study with corresponding fMRI studies that can help identify specific regions and networks that correspond to these whole-brain MEG results.

Are neural representations of emotions relatively consistent across time and situations, or do they change flexibly to adjust to the current milieu? Neurobiological emotion theories differ widely on their answers to this question. Constructivist theories, which have been highly influential in recent theoretical debates over the nature of emotion, posit that emotions are created from combinations of basic cognitive elements (e.g., perception, attention, interoception) shaped by learning and these combinations (called situated conceptualizations) are reactivated when similar situations are encountered again (Barrett, 2014). Lebois et al. (2018) conducted a fMRI study in which subjects were trained to immerse themselves in two different types of situations relating to fear or anger (e.g., physical vs. social threat). During a subsequent test phase, brain activation elicited while subjects generated fear or anger experiences differed considerably depending on whether the learning phase had focused on physical vs. social emotional experiences, thus showing learning effects on the neural correlates of elicited emotional experiences, as assessed by spatial overlap in brain activation. These findings are consistent with the constructivist viewpoint that emotion representations are highly malleable and influenced by learning, just as other, non-affective conceptual representations are. Among the many questions raised by this provocative study are the long-term effects of learned alterations in emotion-related responses and the extent to which multivariate approaches such as the representational similarity analysis used by Grootswagers et al. can further understanding of how learning influences emotion.

Given the wealth of evidence that emotion processing typically involves multiple brain regions acting in concert, there have been several recent efforts to use multivariate fMRI techniques to characterize and classify distributed patterns of brain activity that are associated with different emotion states, such as basic emotions and negative affect (Chang et al., 2015; Kassam et al., 2013; Kragel and LaBar, 2016; Saarimäki et al., 2018). Similar methods have been used to develop

distributed neural markers associated with constructs such as levels of experienced physical pain, as well as vicarious (empathic) pain in which subjects observe pictures of strangers in pain (Krishnan et al., 2016). Lopez-Sola et al. (2017) studied a social emotion, vicarious pain (sometimes called ‘pain empathy’). They asked whether the brain systems that encode vicarious pain are shared with or distinct from those that encode somatic pain. Their similarity or dissimilarity bears on the question of how physically painful and emotionally ‘painful’ experiences are related to one another, and whether empathy engages brain systems that evolved to represent bodily harm to oneself. Previous studies had identified a multivariate brain ‘signature’ that encodes the intensity of somatic pain (Wager et al., 2013), but does not respond to vicarious pain elicited by pictures of pain in others (Krishnan et al., 2016). Conversely, a separate ‘signature’ encoded the intensity of vicarious pain, but showed no response to painful stimuli (Krishnan et al., 2016).

Lopez-Sola et al. provided a direct replication of this double dissociation, but using a different task context. Instead of pictures, they used a ‘cued empathy’ paradigm. Women were scanned with fMRI while their romantic partners remained outside the scanner, visible to the participants through an in-scanner mirror. In some trials, women experienced thermal pain, and in others, they were informed that their partner would be experiencing thermal pain. They applied the identical multivariate brain measures used previously, and found the same pattern: The physical pain ‘signature’ responded only to pain, and the vicarious pain ‘signature’ responded only to observed pain in others. Comparing these brain responses discriminated self-vs. observed pain with 97% accuracy. This study provided a rare example of the use of precisely specified brain markers to test generalization across task variants, and strong evidence that self-pain and pain empathy engage distinct brain processes. The authors were careful to qualify their findings, pointing out that the distributed neural marker of vicarious, empathic pain only captures some of the components that are related to empathy for pain, and that the measure is not a complete or pure measure. The authors suggest several intriguing avenues for further exploration, including clinical applications to conditions involving blunted empathy, and to other constructs related to empathy such as compassion. This study is an excellent example of how recent advances in fMRI analysis have enabled researchers to better understand social-affective constructs such as emotion and empathy by investigating complex, distributed patterns of brain activation.

2. Emotion-cognition interactions

Emotion and cognition have historically been conceived of as separate mental faculties and studied relatively independently. More recently, interest in the ways in which emotion and cognition interact has grown substantially. Understanding the relationships between emotion and cognition, and how these relate to brain function, is a key goal of affective neuroscience. Contemporary theories vary widely on fundamental issues such as the degree to which the neural basis of emotion and cognition are overlapping or separate. Two important types of cognition-emotion interactions represented in this issue are the effects of cognitive control on emotional responses (cognitive reappraisal) and the usually impairing effects of emotion on ongoing cognitive processing.

One of the most well-studied forms of emotion-cognition interactions is emotion regulation, in which individuals change their emotional responses to fit with their goals. There are many forms of emotion regulation. One key distinction is between explicit, intentional emotion regulation and implicit forms of emotion regulation such as divided attention, in which emotional responses are reduced due to decreased processing resources. Using fMRI, Fitzgerald et al. (2018) examined the extent to which an explicit regulation strategy (reappraisal to down-regulate negative emotion) and an implicit regulation manipulation (high vs. low attentional load) engage similar or independent brain regions. The somewhat surprising main finding was that reappraisal and

divided attention engaged independent, non-overlapping brain regions, primarily in prefrontal cortex and anterior cingulate cortex. This finding is surprising because implicit and explicit emotion regulation are generally proposed to share some common processes, such as the deployment of attention. For example, an individual might reappraise a picture of a hand with a bloody cut as being unreal, the hand of an actor in a movie. Reappraisal requires attention and cognitive resources, and thus its underlying components would be expected to overlap partly with a divided attention task. These findings suggest that the neural mechanisms of at least some forms of implicit and explicit regulation are largely distinct. A next step in this line of research will be to examine a wider range of implicit and explicit regulation tasks to assess whether these findings hold more generally.

Emotion regulation studies typically employ a relatively narrow range of emotion-eliciting stimuli, such as pictures or short films. An important question is whether findings generalize to more ecologically valid stimuli and situations. [Speed et al. \(2017\)](#) used ERPs to examine reappraisal of emotions elicited during the retrieval of autobiographical memories of emotional life events. Matching prior findings with laboratory stimuli, recall of emotional memories potentiated the LPP component associated with sustained attention and elaboration, and most importantly, down-regulation using reappraisal reduced the LPP. These findings dovetail with those of a few prior fMRI studies and illustrate how neuroscience studies are increasingly using more ecologically valid stimuli.

Three studies of emotion-cognition interactions focused on interference effects between emotion and cognition and factors that moderate these effects. [Bretherton et al. \(2017\)](#) used ERPs to examine the conditions under which task-irrelevant emotional distractors can be suppressed. A key finding across three separate experiments was that it was much more difficult to suppress the initial shift of attention (as assessed by ERPs) to motivationally salient, threatening human faces than to neutral distractors (scrambled faces). Additionally, the attention-attracting effects of threatening faces were stronger when cognitive resources were decreased by secondary task demands. [Kohn and Fernandez \(2017\)](#) used fMRI to examine the effect of stress and stimulus factors (valence and actor sex) on brain responses during an emotional Stroop task in a large group of men. Their complex design generated a rich and complex set of brain results. Behaviorally, fearful faces and male faces elicited more Stroop interference than happy and female faces. These behavioral differences were also related to activation differences for incongruent trials in key regions previously implicated in cognitive interference tasks. A major effect of stress was to decrease these interference-related brain effects, potentially due to activation of stress-congruent schemas or general alerting effects of stress. The influence of long-lasting emotion-induced states such as stress on emotion-cognition interactions has been relatively understudied and is a promising direction for future research.

In another study of emotional interference, [Réveillon et al. \(2018\)](#) used fMRI to examine the effects of happy, angry, and neutral faces on performance and brain activation in a flanker task in a group of children aged 9–11 years. Although there was no effect of emotional (vs. neutral) stimuli, all faces vs. scrambled faces activated the bilateral amygdala and ventral anterior cingulate cortex. Moreover, this activation was correlated across subjects with faster reaction time to faces when they were congruent stimuli in the flanker task, though there was no relationship between the amygdala and the indices of emotional conflict. The absence of brain effects related to facial emotion in this study is interesting, given the robust effects of facial emotion often seen in studies of adults. Direct comparison across individuals ranging from childhood to adulthood using a similar paradigm may help clarify the developmental timecourse of emotion and emotion-cognition interactions.

Individuals can regulate their affective responses and behavioral reactions to emotional stimuli and situations, typically to strategically change their emotions and mood to align with their goals. For example,

reinterpreting a negative situation as being unimportant or a necessary learning experience can reduce the negative affect it elicits. Although most research has focused on strategic, explicit emotion regulation, emotion regulation can also be induced implicitly, for example, by priming concepts associated with emotional control by presenting the word “restrains” before a target affective stimulus. [Katsumi & Dolcos \(2018\)](#) used fMRI to examine the neural and behavioral effects of explicit and explicit suppression of emotional responses to negative emotional pictures. Behavioral effects included self-reported emotional reactions and subsequent recognition memory. Episodic memory for emotional stimuli is typically enhanced relative to neutral stimuli, and a key mechanism for this effect is increased activation in the amygdala, which in turn enhances memory encoding related activity in the hippocampus and related structures (see also [Inman et al., 2018](#)). The key issue addressed in this study was whether emotion regulation would decrease the emotional enhancement of memory, while at the same time, decreasing amygdala activation and amygdalo-hippocampal connectivity. Both implicit and explicit emotion regulation decreased subsequent episodic memory for the negative pictures, but only explicit regulation was associated with both decreased activity in the amygdala and inferior frontal gyrus (IFG) during encoding and a decreased relationship between activity in these regions and subsequent memory. The neural effects of implicit regulation were less extensive than for explicit regulation, and were primarily observed for decreased functional connectivity between the hippocampus and, separately, the amygdala and IFG. This study makes an important new bridge between the emotion regulation literature and the emotional memory literature, and helps to clarify the similarities and differences between the neural effects of implicit and explicit emotion regulation.

The distinction between memory for items vs. their source (i.e., associated context, such as where one learned a fact) is an important one in the neuroscience of memory, and there is considerable interest in the differential effects of emotion on memory for items vs. their associated source contexts. [Bort et al. \(2017\)](#) used ERPs to examine the neural correlates of item and source memory in a task where neutral items were presented together at encoding with neutral or emotional contexts and memory was tested a week later in a combined recognition and source context memory retrieval task. The main findings were that memory for source context was enhanced when the context was emotional, and behavioral and ERP evidence converged to show that item memory performance was driven by familiarity whereas source memory performance (especially for emotional contexts) was driven by recollection. The timing and location of the ERP component associated with source retrieval resembled the same LPP component that has been strongly linked to emotional memory enhancement and enhanced motivational value and sustained cognitive processing. These findings add to understanding of how different types of memory retrieval processes contribute to item and source memory for emotional stimuli.

Emotion has a wide array of both enhancing and impairing effects on cognition. A key area of interest has been the degree to which task-irrelevant emotional stimuli can impair cognitive task processing by capturing and diverting processing resources such as attention. For example, task-irrelevant highly negative emotional pictures presented during the maintenance phase of a working memory task can impair performance ([Dolcos and McCarthy, 2006](#)). This impairing effect of emotion on working memory has been attributed to reciprocal inhibition between a so-called “cold” dorsal executive system network (dorsolateral PFC and lateral parietal cortex) and a “hot” emotional processing network involving the VMPFC and amygdala.

In line with this previous literature, [Langeslag and van Strien \(2017\)](#) note that romantic love can have either enhancing or impairing effects on cognition. For example, daydreaming about one’s beloved can be distracting and can interfere with cognitive task performance. In two ERP experiments, subjects who had recently fallen in love performed a visuospatial working memory task and the faces of either their newly-beloved, a close friend, or a stranger were presented as

task-irrelevant stimuli either during the maintenance or the encoding phase. As predicted, the late positive potential (LPP), related to sustained motivated attention was largest for beloved faces (especially those rated higher in arousal), though other components did not differ, suggesting that the effect of romantic love enhanced motivated attention but not early components related to perceptual processing. Although none of the task-irrelevant distractor types impaired task performance, perhaps due to issues related to task difficulty, there was a marginally significant correlation between obsessive thoughts about the beloved and poorer performance on the task. These findings suggest that highly arousing and motivating social stimuli (one's beloved) can influence brain activity even when they are task irrelevant, and that this effect is on an ERP component linked to sustained attention.

3. Individual differences in emotion

Several papers in this issue investigate the role of individual differences in emotion and related constructs such as anxiety and personality. Three of these papers focus on brain structure whereas two focus on functional brain measures (fMRI and ERP).

Neuropsychological studies of the effects of brain lesions on personality and emotionality have overwhelmingly focused on negative, deleterious effects, particularly with respect to prefrontal lesions. A wide range of negative effects have been reported, including poor emotion regulation, labile affect, aggression, apathy, blunted subjective emotional experience and insight, and increased anxiety and depression. In their intriguing report, [King et al. \(2017\)](#) look for the silver lining in this cloud of impairments. They note that clinical observations have occasionally reported improvements in personality and emotional behavior in the wake of focal brain lesions. Taking advantage of a large group of well-characterized patients, they examined regional associations between brain areas and changes in personality and emotionality to determine the frequency and characteristics of post-lesion improvements and the neuroanatomical correlates related to improvement in these measures.

Combining group results with illustrative single case studies, a clear picture emerged. Lesions to the bilateral frontal polar regions and right anterior dorsolateral prefrontal region were most strongly associated with improvements in personality, and these improvements were more marked in patients who had more severe disturbance of personality and emotion before their lesions (no patients had premorbid psychiatric illness). Though exploratory, these findings fit well with other evidence that altering dysfunctional or overactive brain circuits, for example, with deep brain stimulation, can ameliorate symptoms of depression and obsessive-compulsive disorder ([Riva-Posse et al., 2019](#); [Vicheva et al., 2020](#)). The careful characterization of positive changes in personality and emotion associated with focal brain lesions in specific brain regions sets the stage for future studies that can explore other aspects of emotion and social function such as empathy. Another promising direction for further study is using fMRI to assess how focal lesions alter resting state brain networks and task-based fMRI brain responses are associated with improved or enhanced aspects of personality and emotion.

[Hu et al. \(2017\)](#) took a different approach to examining relationships between brain structure and emotion by exploring brain-behavior relationships in a healthy, nonclinical sample. Using structural MRI, they assessed whether variations in the size the amygdala, a key structure involved in affective processing, are associated with trait anxiety, and whether the personality trait of neuroticism (associated with a greater tendency towards negative affect) might mediate this relationship. Smaller amygdala volume has been previously found to be associated with depression, panic disorders, and social anxiety disorder, but whether decreased amygdala volume represents a predisposing risk factor for anxiety disorders or is a consequence of such disorders remains unclear. Consistent with a risk factor view, decreased left amygdala volume was significantly associated with trait anxiety, and this relationship was mediated by neuroticism. This finding nicely connects prior

work with clinical populations to normal variation in amygdala size, and adds to the wealth of evidence pointing to the amygdala as a key structure mediating variations in emotion, mood, and personality. A key goal for further study is to build a detailed longitudinal account of how this and other structural risk factors contribute to the development of affective disorders.

The third study focusing on structural brain variation and emotion, [Coad et al. \(2017\)](#), examined the relationship between variations in structural connectivity in a key white matter tract, the uncinate fasciculus (UF) and the ability to decode facial emotion. The UF connects the orbital and medial prefrontal cortex with anterior temporal lobe regions such as the amygdala and temporal pole, regions that comprise a proposed system that mediates social and emotional behavior, sometimes referred to as the “anterior temporal system” ([Inhoff and Ranganath, 2017](#)). Disruption of the UF has been implicated in several clinical disorders characterized by altered social behavior. Using a similar logic as Hu et al. motivated by these clinical associations with UF disruption, Coad et al. examined whether variation in UF microstructural integrity (fractional anisotropy; FA) in a healthy population was associated with performance on tasks assessing decoding of facial emotion. The results of two separate experiments showed that decreased right UF microstructural integrity was associated with poorer performance on both the “reading of the mind in the eyes” task (requiring the decoding of subtle emotional cues from the eyes alone) and a standard facial emotion recognition task. Interestingly, these findings were specific to the right UF and to emotion processing, as facial identity processing ability was not related to FA integrity. These findings shed light on the functional role of the UF, which to date has been relatively unclear, yet many questions remain, for example, regarding the developmental trajectory of the UF and how individual differences in UF integrity relate to functional connectivity measures such as resting state connectivity within the anterior temporal system.

Two studies examined the role of individual differences in ERP and emotion, both focusing on a specific ERP component, the Late Positive Potential (LPP), associated with increased capture of attentional and cognitive resources by motivationally salient stimuli such as emotional pictures. Using a novel application of sophisticated single-trial-level ERP analysis approach, [Stange et al. \(2017\)](#) examined the relationship between ERP and behavioral responses to task-irrelevant emotional and neutral pictures and the degree of treatment response to cognitive behavioral therapy (CBT) in a group of patients with anxiety and/or depression. Based on prior evidence that greater cognitive flexibility is associated with better treatment outcome, Stange et al. focused on whether a measure of increased cognitive-emotional flexibility (referred to as brain-behavior coupling) would predict CBT treatment outcome. Prior to treatment, patients carried out a speeded reaction time task in which task-irrelevant stimuli (either aversive or neutral pictures) were presented concurrently while EEG was recorded. As expected, task aversive pictures captured attention more than neutral pictures, reflected by greater LPPs and slowed RTs on the subsequent trial. The brain-behavior flexibility measure assessed whether an individual benefited behaviorally (faster RTs) from lower LPPs on the previous trial, thus demonstrating greater behavioral adaptability. The key finding was that patients who showed greater flexibility on the brain-behavioral coupling measure were more likely to respond to treatment and show improvement in their depressive symptoms. This study is an excellent illustration of how basic findings from studies of healthy individuals (LPP and RT effects of aversive stimuli) can be used to probe processes contributing to clinical disorders. Such studies both contribute to understanding of basic cognitive and emotional mechanisms underlying clinical disorders and point to ways that affective neuroscience measures can be used to help tailor the selection of treatments to individuals to maximize treatment efficacy.

[Hartigan and Richards, 2017](#) examined the degree to which a neural measure of emotion processing (ERP components) to disgusting, threatening, or neutral picture stimuli were influenced by disgust

propensity, disgust sensitivity and attentional control following exposure to disgusting information, in a group of healthy control subjects. As was found by Stange et al. LPPs were increased for the emotional stimuli (relative to neutral stimuli). Interestingly, individual differences in disgust propensity were associated with reduced LPPs to emotional stimuli, though this effect was only seen for subjects who had been previously exposed to disgust-related information. This study illustrates the complex interplay between individual differences, attentional control, and brain responses to emotional stimuli. It further shows how different types of negative emotional stimuli, which sometimes are treated as functionally equivalent in affective neuroscience studies, can behave quite differently depending on the experimental task demands.

4. Conclusion and future directions

We hope that findings and theoretical concepts in this special issue will spark interest into new research directions in affective neuroscience. Indeed, some papers, such as Dubois et al. explicitly present a novel research framework and outline promising future research questions. Where relevant, we have also suggested some avenues for further research. The field of affective neuroscience continues to expand and to incorporate more sophisticated methodologies and analysis methods in the effort to better understand both the local and distributed brain processes underlying emotion. With increasing use of methods that can establish causality, such as intracranial electrical brain stimulation and TMS/tDCS, causal hypotheses that have so far been explored primarily in nonhuman animals and neuropsychological patients with focal brain lesions can be explored using more powerful and flexible tools.

In conclusion, the papers in this special issue illustrate a diverse group of exciting lines of research into the neural basis of emotion and related topics. We hope that the novel findings and theoretical ideas from these studies will foster new research into these issues by a wide range of basic and clinical affective neuroscience researchers. Some of the greatest challenges for the future will be to integrate findings across different methods and levels of analysis from the single neuron to the whole brain. A related challenge will be to synthesize novel findings to revise current neurobiological theories of emotion and emotion-cognition interactions, providing new insights into the neural basis of emotion.

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Stephan Hamann^{a,*}, Luiz Pessoa^b, Tor D. Wager^c

^a Department of Psychology, Emory University, USA

^b Department of Psychology, University of Maryland, College Park, USA

^c Department of Psychological & Brain Sciences, Dartmouth College, USA

* Corresponding author.

E-mail address: shamann@emory.edu (S. Hamann).